Classical Field Theory — Exercise Sheet 12

Completing transverse reduction

Hints and answers

Use the unitary Fourier transform of Lecture 4. The real canonical form is

$$\Omega((a,p),(b,q)) = \int (a\cdot q - b\cdot p).$$

The projector is $P_{\mathrm{T}}(k) = I - kk^{\mathsf{T}}/|k|^2$ for $k \neq 0$. All claims about a fixed annulus below concern the data and smearings specified in Lecture 12, not pointwise Fourier observables.

Exercise 12.1 (A finite Dirac bracket and an oblique mode). For a fixed $k \in \mathbb{R}^3 \setminus \{0\}$, consider the finite canonical model $(a,p) \in \mathbb{R}^3 \times \mathbb{R}^3$ with $\{a_i, p_j\} = \delta_{ij}$. Set $c_1 = k\cdot p$ and $c_2 = k\cdot a$. Given an invertible constraint matrix $C_{ab} = \{c_a, c_b\}$, its *Dirac bracket* is

$$\{F,G\}_D = \{F,G\} - \{F,c_a\}(C^{-1})_{ab}\{c_b,G\}.$$

This formula is supplied for this finite system.

(a) Compute C, its inverse, and every coordinate Dirac bracket. Check that each constraint has zero Dirac bracket with every coordinate.

(b) At $k = (1,2,2)$, display the projector and project $(3,-1,2)$. Recover the annulus smeared bracket of Lecture 12 from this matrix symbol, using real vector smearings f, g.

(c) At $k = 0$, which step fails? Explain why periodic real gauge parameters on $\mathbb{T}^3$ leave three constant canonical pairs, and give their form.

Hint. Compute the two off-diagonal constraint brackets before inverting. For smearings use symmetry and idempotence of the Fourier multiplier. A periodic gradient has zero mean.

Answer. The nonzero bracket is $\{a_i, p_j\}_D = P_{\mathrm{T}}(k)_{ij}$. At $(1,2,2)$ the matrix and projection are

$$\frac{1}{9}\begin{pmatrix} 8 & -2 & -2 \\ -2 & 5 & -4 \\ -2 & -4 & 5 \end{pmatrix}, \qquad (22,-19,8)/9.$$

The zero-mode form is $(2\pi)^3 \sum_i \delta a_i \wedge \delta p_i$.

Exercise 12.2 (A union of annuli, rather than one annulus). For $n \geq 2$ let $\mathcal{U}_n = \{n^{-1} < |k| < n\}$ and let $\mathcal{Y}$ be the union of the real Schwartz classes with smooth compact Fourier support in $\mathcal{U}_n$.

(a) Prove that $\mathcal{Y}$ is dense in the real spaces $L^2(\mathbb{R}^3;\mathbb{R})$ and $H^1(\mathbb{R}^3;\mathbb{R})$, using the appropriate norm in each case. You may use the usual L^2 density of compact smooth functions, proved by truncation and smoothing.

(b) Prove the corresponding density for transverse vector fields, with approximants $P_{\mathrm{T}}v_n$. Keep the real-field Fourier condition.

(c) Show that the analogous class in one fixed annulus is not dense in either whole space. Explain why a bounded extension from a dense union must not be justified by density of just one annulus.

Hint. Cut off low and high frequencies, smooth inside the resulting annulus, and average a profile with its conjugate reflection. Fourier weights are bounded above and below on each compact annulus.

Answer. The union is dense in both norms, also after transverse projection. One fixed annulus misses every nonzero profile supported in a disjoint frequency shell and therefore is not dense.

Exercise 12.3 (A gauge parameter outside L^2). Choose a real radial $\eta \in C_c^\infty(\mathbb{R}^3)$ equal to one on $|k| \le 1$. Define a tempered scalar distribution and a vector by their Fourier profiles

$$\widehat{\chi}_0(k) = \frac{\eta(k)}{|k|^2}, \qquad \widehat{A}(k) = ik\,\widehat{\chi}_0(k) \quad (k \neq 0).$$

A locally integrable function of at most polynomial growth defines a tempered distribution. You may also use that a distribution with all first derivatives zero is constant.

(a) Show that χ_0 is well-defined and real, that $A \in H^1$, and that $\chi_0 \notin L^2$. Can any L^2 scalar have gradient A?

(b) For a general real $A \in H^1(\mathbb{R}^3;\mathbb{R}^3)$ compare the two profiles $\widehat{\chi}_L = -i(k\cdot\widehat{A})/|k|^2$ and $-\widehat{\chi}_L$. Which one has gradient $(I - P_{\mathrm{T}})A$, and which gauges A to $P_{\mathrm{T}}A$? Justify that χ_L is tempered even near zero frequency.

Hint. Use the radial measure $r^2 \mathrm{d}r\mathrm{d}\Omega$ when testing integrability. For the general parameter use Cauchy–Schwarz with $|k|^{-1}$ near zero, and test against Schwartz functions at infinity.

Answer. The gradient A belongs to H^1 although its parameter has no L^2 representative, even modulo constants. The parameter χ_L represents the longitudinal part; the gauge transformation uses $-\chi_L$.

Exercise 12.4 (Weak symplectic does not mean every gradient exists). On the real space $\mathcal{E}_T = H^1_{\mathrm{trans}} \times L^2_{\mathrm{trans}}$, use the canonical form from Lecture 12. Let $\theta(r)$ be smooth, zero for $r \le 1$ and one for $r \ge 2$, and define the real transverse L^2 vector h by

$$\widehat{h}(k) = \theta(|k|)|k|^{-2}P_{\mathrm{T}}(k)e_1.$$

(a) Prove that $h \in L^2$ but $h \notin H^1$. Show that $F(a,p) = \int h \cdot p\,\mathrm{d}^3x$ is continuous on $\mathcal{E}_T$.

(b) Show that no $X \in \mathcal{E}_T$ satisfies $\iota_X\Omega = DF$. Why does this not contradict the nondegeneracy proved in Lecture 12?

(c) Does finiteness of $\int(|\operatorname{curl} A|^2 + |\Pi|^2)$ imply membership in this inhomogeneous phase space? Test a nonzero constant potential with zero momentum.

Hint. The angular integral of $|P_{\mathrm{T}}(n)e_1|^2$ is $8\pi/3$. Test the proposed Hamiltonian vector against arbitrary transverse momentum variations before considering its other component.

Answer. The high-frequency radial integrals are proportional to $\int_2^\infty r^{-2}\mathrm{d}r$ for L^2 and $\int_2^\infty (1+r^2)r^{-2}\mathrm{d}r$ for H^1. A representing vector would require $X_A = h$, impossible in H^1. Finite curl energy alone does not imply $A \in L^2$.